

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG



BÁO CÁO BÀI TẬP 7/4

Họ Và Tên: Nguyễn Quốc Anh

Mã Số Sinh Viên: 20235263

Học Phần: IT4060

Mã Lớp: 168503

Github: <https://github.com/quocanh13/IT4060>

Bài 1: Telnet Server

- Chạy server: gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server && ./5-5-ex1-telnet_server

The screenshot shows a VS Code editor with a C program for a Telnet server. The code includes standard headers, defines a server IP and port, and implements a main function that listens for connections and prints a message. The terminal shows the successful compilation of the program into an executable named 5-5-ex1-telnet_server and its execution, resulting in the output 'Server is listening'.

```
1 #include <stdio.h>
2
3 // gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server && ./5-5-ex1-telnet_server
4
5 #include <sys/socket.h>
6 #include <arpa/inet.h>
7 #include <netinet/in.h>
8 #include <unistd.h>
9 #include <sys/select.h>
10 #include <signal.h>
11
12
13 int check_account(char name[], int len);
14 void execute_command(char command[], int len);
15
16 int client;
17 int server;
18 int verified = 0;
19
20 int main(){
21     char *server_ip = "127.0.0.1";
22     int server_port = 5000;
23     struct sockaddr_in server_addr = {0};
24
25     server_addr.sin_family = AF_INET;
26     server_addr.sin_port = htons(server_port);
27     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
28
29     server = socket(AF_INET, SOCK_STREAM, 0);
30
31     if(server < 0){
32         printf("Failed to create server socket\n");
33         exit(0);
34     }
35     if(bind(server, (struct sockaddr*)&server_addr, sizeof(server_addr)) < 0) {
36         printf("Failed to bind server\n");
37         exit(0);
38     }
39     if(listen(server, 10) < 0) {
40         printf("Failed to listen\n");
41     }
42 }
```

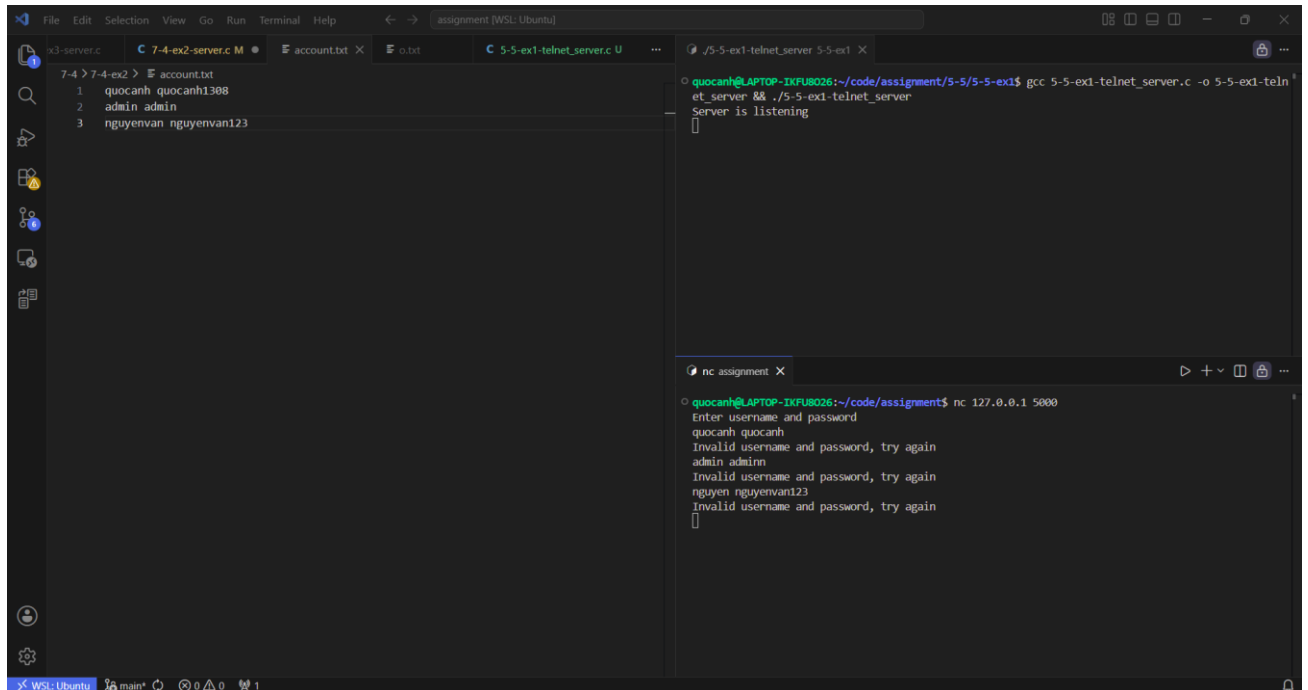
```
quocanh@LAPTOP-IKFU8026:~/code/assignment/5-5/5-5-ex1$ gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server && ./5-5-ex1-telnet_server
Server is listening
```

- Dùng netcat tạo kết nối client đến server: nc 127.0.0.1 5000
- Server sẽ gửi yêu cầu nhập username và password

The screenshot shows the netcat client connecting to the server. The terminal displays the netcat command and the subsequent connection. The server's output 'Server is listening' is visible. The netcat client's output shows the connection established and the prompt 'Enter username and password'.

```
quocanh@LAPTOP-IKFU8026:~/code/assignment/5-5/5-5-ex1$ nc 127.0.0.1 5000
Enter username and password
```

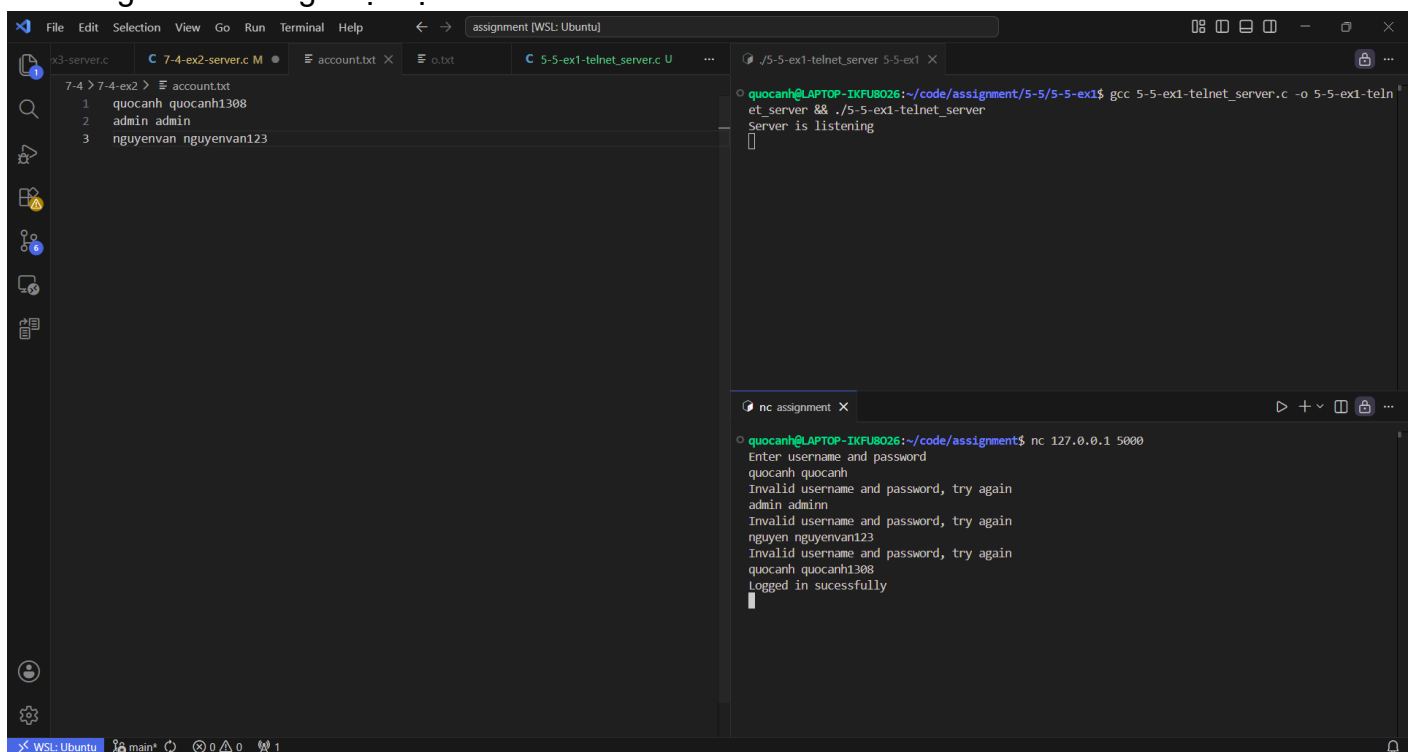
- Người dùng nhập sai username và password thì server yêu cầu nhập lại



```
quocanh@LAPTOP-IKFU8026:~/code/assignment/5-5/5-5-ex1$ gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server
Server is listening

nc assignment
quocanh@LAPTOP-IKFU8026:~/code/assignment$ nc 127.0.0.1 5000
Enter username and password
quocanh quocanh
Invalid username and password, try again
admin admin
Invalid username and password, try again
nguyen nguyen123
Invalid username and password, try again
```

- Khi client nhập đúng username và password thì server sẽ thông báo đăng nhập thành công và sẵn sàng nhận lệnh



```
quocanh@LAPTOP-IKFU8026:~/code/assignment/5-5/5-5-ex1$ gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server
Server is listening

nc assignment
quocanh@LAPTOP-IKFU8026:~/code/assignment$ nc 127.0.0.1 5000
Enter username and password
quocanh quocanh
Invalid username and password, try again
admin admin
Invalid username and password, try again
nguyen nguyen123
Invalid username and password, try again
quocanh quocanh1308
logged in successfully
```

- Client gửi lệnh ls -l thì phía server thực thi lệnh, ghi kết quả ra o.txt và gửi cho client

```
File Edit Selection View Go Run Terminal Help
assignment [WSL: Ubuntu]

C 28-4-ex3-server.c o.txt U X
5-5 > 5-5-ex1 > o.txt
1 total 28
2 -rwxr-xr-x 1 quocanh quocanh 17032 May 11 03:58 5-5-ex1-telnet_server
3 -rw-r--r-- 1 quocanh quocanh 3409 May 11 03:35 5-5-ex1-telnet_server.c
4 -rw-r--r-- 1 quocanh quocanh 54 May 11 03:16 account.txt
5 -rw-r--r-- 1 quocanh quocanh 0 May 11 04:05 o.txt
6

o quocanh@LAPTOP-1KFU8026:~/code/assignment/5-5/5-5-ex1$ gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server
Server is listening

nc assignment
o quocanh@LAPTOP-1KFU8026:~/code/assignment$ nc 127.0.0.1 5000
Enter username and password
quocanh quocanh
Invalid username and password, try again
admin adminn
Invalid username and password, try again
nguyen nguyenvan123
Invalid username and password, try again
quocanh quocanh1308
Logged in successfully
ls -l
total 28
-rwxr-xr-x 1 quocanh quocanh 17032 May 11 03:58 5-5-ex1-telnet_server
-rw-r--r-- 1 quocanh quocanh 3409 May 11 03:35 5-5-ex1-telnet_server.c
-rw-r--r-- 1 quocanh quocanh 54 May 11 03:16 account.txt
-rw-r--r-- 1 quocanh quocanh 0 May 11 04:05 o.txt
```

- Ví dụ với lệnh pwd

```
File Edit Selection View Go Run Terminal Help
assignment [WSL: Ubuntu]

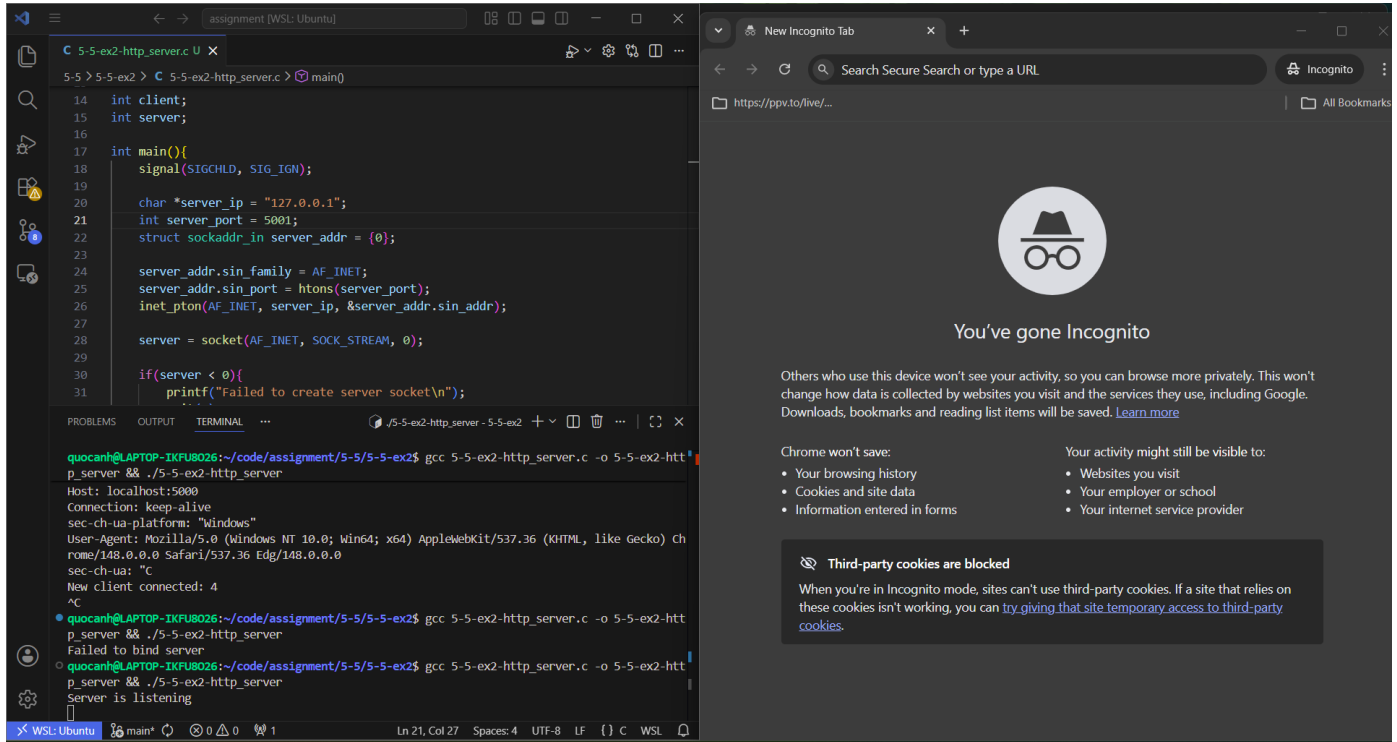
C 28-4-ex3-server.c o.txt U X
5-5 > 5-5-ex1 > o.txt
1 /home/quocanh/code/assignment/5-5/5-5-ex1
2

o quocanh@LAPTOP-1KFU8026:~/code/assignment/5-5/5-5-ex1$ gcc 5-5-ex1-telnet_server.c -o 5-5-ex1-telnet_server
Server is listening

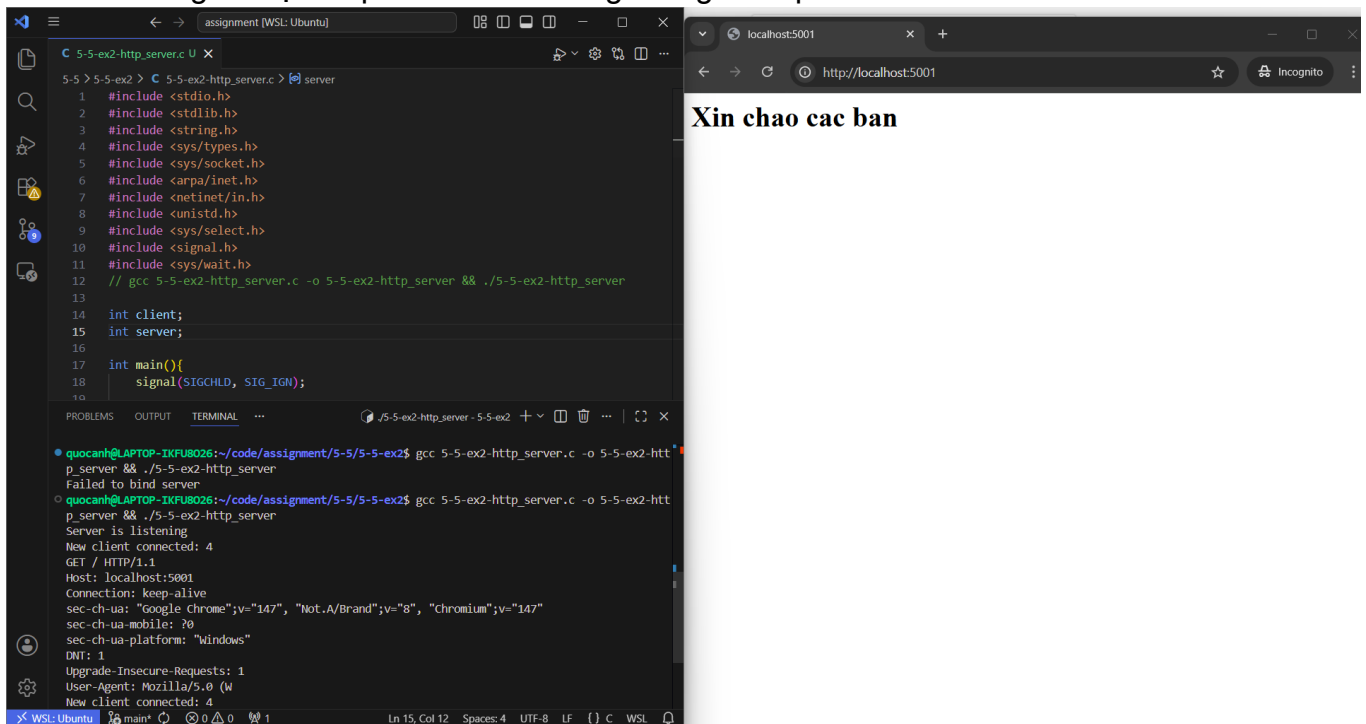
nc assignment
o quocanh@LAPTOP-1KFU8026:~/code/assignment$ nc 127.0.0.1 5000
Enter username and password
quocanh quocanh
Invalid username and password, try again
admin adminn
Invalid username and password, try again
nguyen nguyenvan123
Invalid username and password, try again
quocanh quocanh1308
Logged in successfully
ls -l
total 28
-rwxr-xr-x 1 quocanh quocanh 17032 May 11 03:58 5-5-ex1-telnet_server
-rw-r--r-- 1 quocanh quocanh 3409 May 11 03:35 5-5-ex1-telnet_server.c
-rw-r--r-- 1 quocanh quocanh 54 May 11 03:16 account.txt
-rw-r--r-- 1 quocanh quocanh 0 May 11 04:05 o.txt
pwd
/home/quocanh/code/assignment/5-5/5-5-ex1
```

Bài 2: HTTP Server

- Chạy server: `gcc 5-5-ex2-http_server.c -o 5-5-ex2-http_server && ./5-5-ex2-http_server`

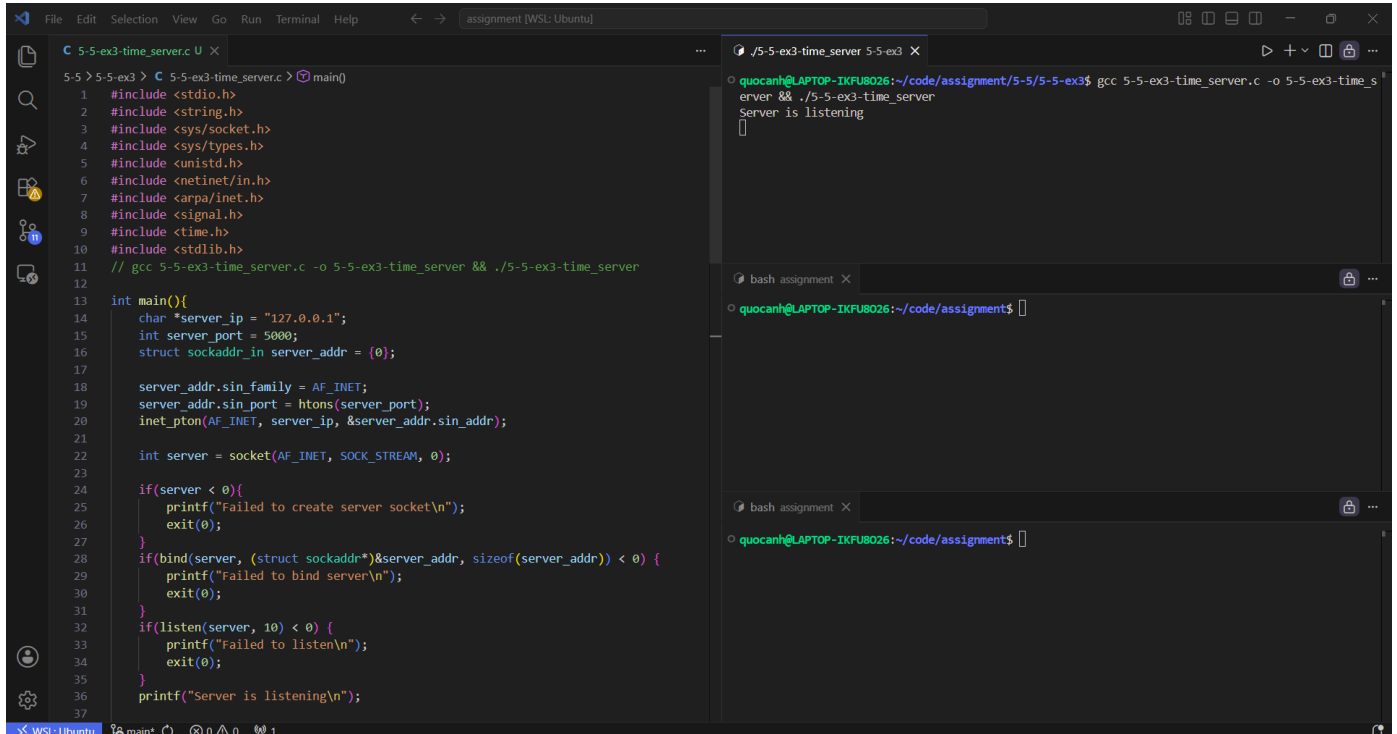


- Truy cập vào server thông qua trình duyệt web với địa chỉ : `http://localhost:5001/`
- Server ghi nhận request và in ra đồng thời gửi response cho client



Bài 3: Time Server

- Chạy server : `gcc 5-5-ex3-time_server.c -o 5-5-ex3-time_server && ./5-5-ex3-time_server`



- Tạo 2 client kết nối đến server
- Client gửi các yêu cầu sai format và bị server từ chối “Invalid Format”

```

C 5-5-ex3-time_server.c U X
1 #include <stdio.h>
2 #include <string.h>
3 #include <sys/socket.h>
4 #include <sys/types.h>
5 #include <unistd.h>
6 #include <netinet/in.h>
7 #include <arpa/inet.h>
8 #include <signal.h>
9 #include <time.h>
10 #include <stdlib.h>
11 // gcc 5-5-ex3-time_server.c -o 5-5-ex3-time_server && ./5-5-ex3-time_server
12
13 int main(){
14     char *server_ip = "127.0.0.1";
15     int server_port = 5000;
16     struct sockaddr_in server_addr = {0};
17
18     server_addr.sin_family = AF_INET;
19     server_addr.sin_port = htons(server_port);
20     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
21
22     int server = socket(AF_INET, SOCK_STREAM, 0);
23
24     if(server < 0){
25         printf("failed to create server socket\n");
26         exit(0);
27     }
28     if(bind(server, (struct sockaddr*)&server_addr, sizeof(server_addr)) < 0) {
29         printf("failed to bind server\n");
30         exit(0);
31     }
32     if(listen(server, 10) < 0) {
33         printf("failed to listen\n");
34         exit(0);
35     }
36     printf("Server is listening\n");
37 }

./5-5-ex3-time_server 5-5-ex3 X
quocanh@LAPTOP-1KFU8026:~/code/assignment/5-5/5-5-ex3$ gcc 5-5-ex3-time_server.c -o 5-5-ex3-time_server && ./5-5-ex3-time_server
Server is listening

nc assignment X
quocanh@LAPTOP-1KFU8026:~/code/assignment$ nc 127.0.0.1 5000
abvdsdf
Invalid Format
GET_TIME dd-mm-yyyy
Invalid Format
GET_TIME mm/yy/dddd
Invalid Format
[]

nc assignment X
quocanh@LAPTOP-1KFU8026:~/code/assignment$ nc 127.0.0.1 5000
get_time dd-mm-yyyy
Invalid Format
get_time dd/mm/yyyy
Invalid Format
[]

```

- Clients gửi đúng format và nhận được kết quả tương ứng

```

C 5-5-ex3-time_server.c U X
1 #include <stdio.h>
2 #include <string.h>
3 #include <sys/socket.h>
4 #include <sys/types.h>
5 #include <unistd.h>
6 #include <netinet/in.h>
7 #include <arpa/inet.h>
8 #include <signal.h>
9 #include <time.h>
10 #include <stdlib.h>
11 // gcc 5-5-ex3-time_server.c -o 5-5-ex3-time_server && ./5-5-ex3-time_server
12
13 int main(){
14     char *server_ip = "127.0.0.1";
15     int server_port = 5000;
16     struct sockaddr_in server_addr = {0};
17
18     server_addr.sin_family = AF_INET;
19     server_addr.sin_port = htons(server_port);
20     inet_pton(AF_INET, server_ip, &server_addr.sin_addr);
21
22     int server = socket(AF_INET, SOCK_STREAM, 0);
23
24     if(server < 0){
25         printf("failed to create server socket\n");
26         exit(0);
27     }
28     if(bind(server, (struct sockaddr*)&server_addr, sizeof(server_addr)) < 0) {
29         printf("failed to bind server\n");
30         exit(0);
31     }
32     if(listen(server, 10) < 0) {
33         printf("failed to listen\n");
34         exit(0);
35     }
36     printf("Server is listening\n");
37 }

./5-5-ex3-time_server 5-5-ex3 X
quocanh@LAPTOP-1KFU8026:~/code/assignment/5-5/5-5-ex3$ gcc 5-5-ex3-time_server.c -o 5-5-ex3-time_server && ./5-5-ex3-time_server
Server is listening

nc assignment X
quocanh@LAPTOP-1KFU8026:~/code/assignment$ nc 127.0.0.1 5000
abvdsdf
Invalid Format
GET_TIME dd-mm-yyyy
Invalid Format
GET_TIME mm/yy/dddd
Invalid Format
GET_TIME dd/mm/yyyy
11/05/2026
GET_TIME yyyy/dd/mm
2026/11/05
[]

nc assignment X
quocanh@LAPTOP-1KFU8026:~/code/assignment$ nc 127.0.0.1 5000
get_time dd-mm-yyyy
Invalid Format
get_time dd/mm/yyyy
Invalid Format
GET_TIME mm/dd/yyyy
05/11/2026
GET_TIME dd/yyyy/mm
11/2026/05
GET_TIME mm/yyyy/dd
05/2026/11
[]

```

